Diploma in Computer Engineering

Temasek Polytechnic | School of Engineering

IIOT for Advanced Manufacturing (ECMC104)

Lab 7 – Raspberry PI GPIO

Objectives

1. To learn how to use the GPIO library for Raspberry Pi.
2. To learn how to connect the GPIO of Raspberry Pi.
3. To learn how to use the **scp** command to transfer a file over the network.

Equipment

1 x LED
1 x Breadboard
1 x 220 to 470 Ω resistor
Jumper wires

Contents

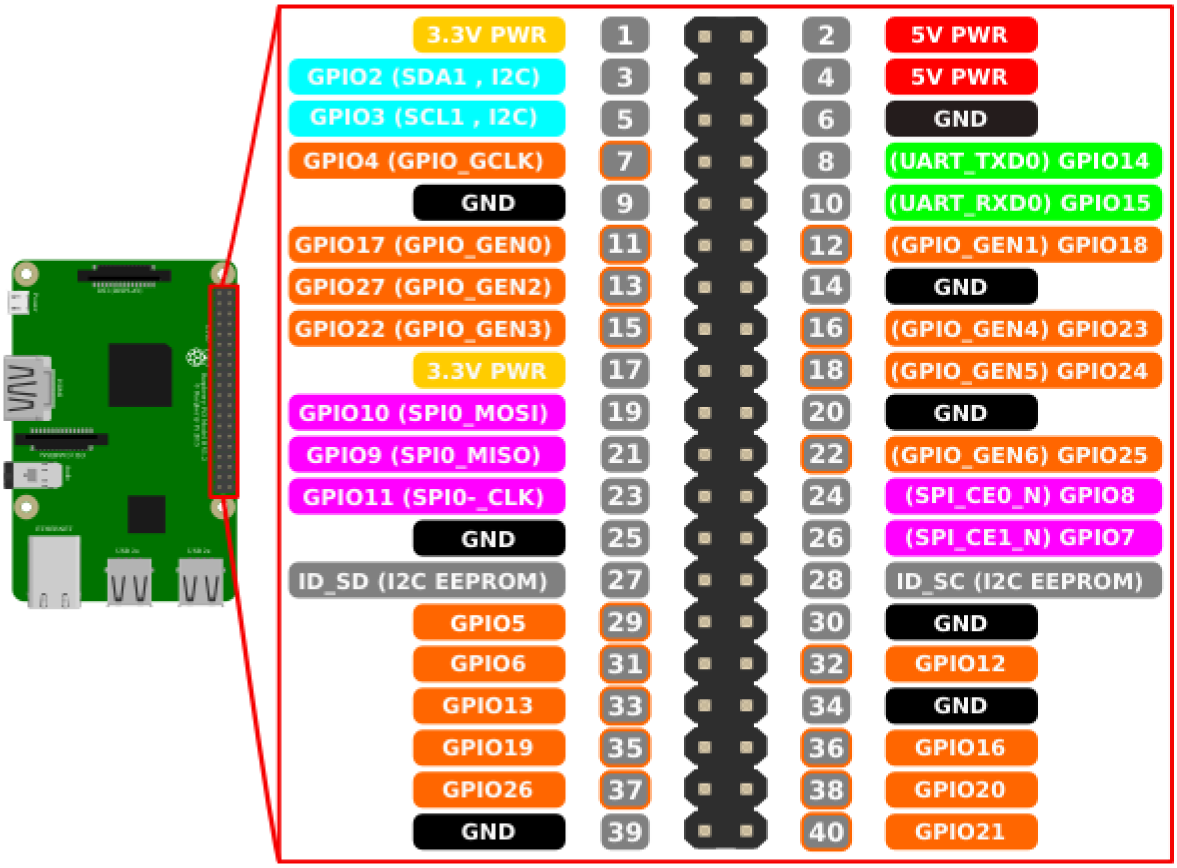
1 Using the GPIO in Raspberry Pi

Introduction

In this exercise, you will learn how to use Raspberry Pi's GPIO. On a breadboard, you will connect an LED. You will then code a program in Python to turn on or off this LED.

1.1 Pin Layout for Raspberry Pi

The following diagram shows Raspberry Pi's pin layout. Take note of the pins which are named as **GPIO**XX. We will be mainly using one of these pins for I/O programming.

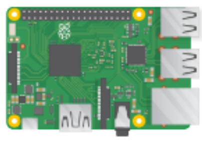


Raspberry Pi Pinout

1.2 Wiring the Pin

Direct Connection

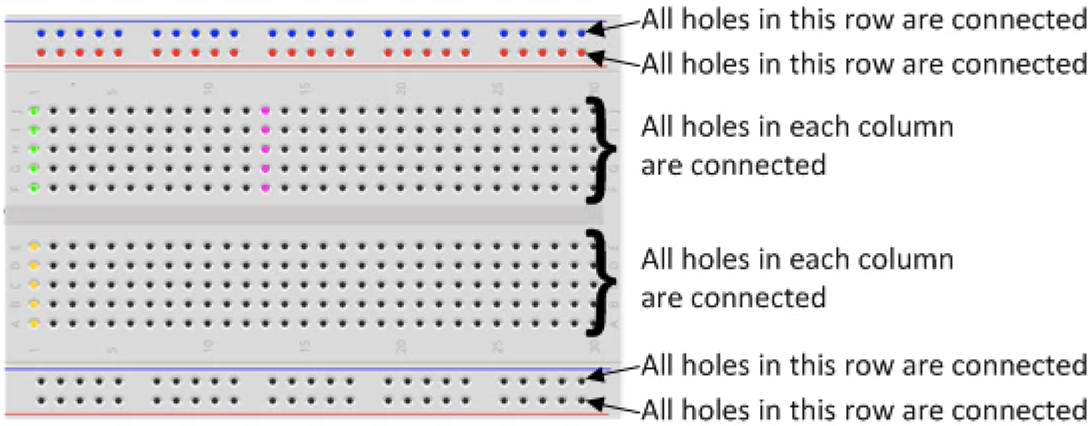
You can use jumper wires to connect directly from the Raspberry Pi board to the breadboard.

Breadboard	Male-to-female jumper wires	LED	Resistor	Power component
				

The Breadboard

The breadboard is a way of connecting electronic components to each other without having to solder them together. They are often used to test a circuit design before creating a Printed Circuit Board (PCB).

The holes on the breadboard are connected in a pattern.



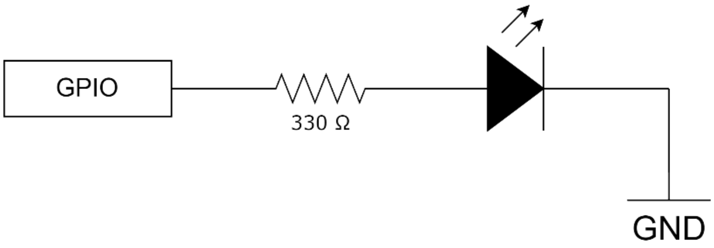
Reference:
Breadboard tutorial: How to use a breadboard (for beginners)
https://www.youtube.com/watch?v=W6mixXsn-Vc

Connecting an LED

Connect an LED to one of the GPIO pin. Note down the GPIO pin you have used.

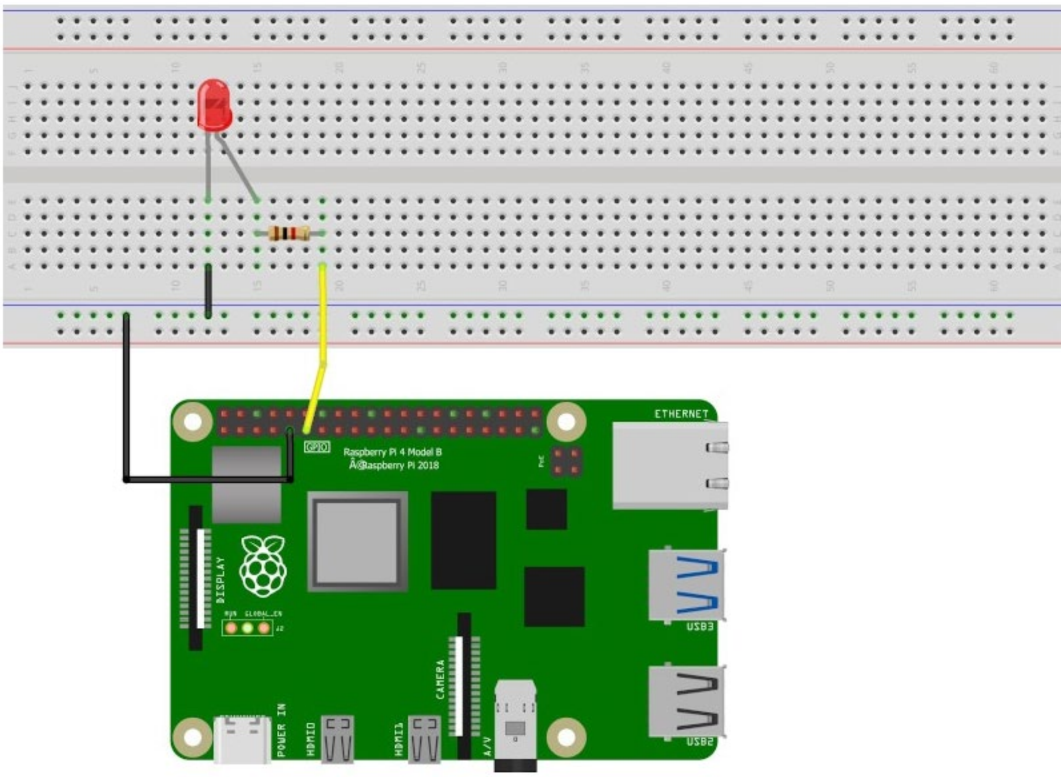
GPIO Pin:	____

Connect a 330 Ω resistor in series with the LED. The full connection is shown as a simple circuit below.



LED Circuit

The schematics to plug an LED to your Raspberry Pi as follow.



1.3 Writing the Python Program in Ubuntu

In WSL2/Ubuntu Terminal, type out the following Python program.

```
import RPi.GPIO as GPIO
import time

GPIO.setmode(GPIO.BCM)
GPIO.setup(17,GPIO.OUT)

while True:
    GPIO.output(17,GPIO.HIGH)
    time.sleep(1)

    GPIO.output(17,GPIO.LOW)
    time.sleep(1)

GPIO.cleanup()
```

simple_io.py

This program uses the **GPIO17** pin to toggle an LED. You can modify the pin according to what you have used.

To turn on and off the LED with a delay interval, use the **sleep()** library function from the **time** module.

The **sleep()** function suspends execution of the current program for a given number of seconds. It takes a floating-point number as an argument.

1.4 Challenge

Now, how can you modified “simple_io.py” to make sure the LED GPIO (and all other GPIOs) is cleaned up every time you kill the Python program?

simple_io_complete.py

1.5 Transferring the Program File to Raspberry Pi

Since the program needs to run from Raspberry Pi, it is necessary to transfer this file to the device.

Connecting to Raspberry Pi from Ubuntu (Laptop)

Follow the procedure in Lab 4 to ensure that your setup enables network connectivity from your laptop to Raspberry Pi.

When it is able to connect to Raspberry Pi, continue with the procedure as follows.

Transferring the File from Ubuntu to Raspberry Pi

Use the **scp** (secure copy) command to copy the Python program that you have written to Raspberry Pi.
Assume that you want to copy it to Pi's **/home/pi/Develop** folder, you need to create the "Develop" folder in the home directory first.

```
$ scp simple_io.py pi@192.168.1.10:/home/pi/Develop
```

Change "**192.168.1.10**" accordingly to your own Raspberry Pi's IP address.
Enter "**raspberry**" as the password when you are prompted.

1.6 Running the Program in Raspberry Pi

Remote login with the **ssh** command:

```
$ ssh pi@192.168.1.10
```

Enter the password again.

Go to the **/home/pi/Develop** folder and list the contents.

```
$ cd /home/pi/Develop
$ ls
```

Run the program:

```
$ python3 simple_io.py
```

Do you observe the LED blinking according to the durations you have specified? Write down your observations below:

Connecting a Buzzer

Patch up the circuit with another GPIO pin to test the buzzer. Note down the GPIO pin you have used.

Demonstrate to your tutor.

GPIO Pin:	____

~ End of Lab 7 ~